Homework 9 – Dynamic Programming

Due: 13:30, December 31, 2025

Note: In this course, we refer to an integer program (IP) as a linear program (LP) with the restriction that certain variables must take integer values. Therefore, in an IP model, the objective function and the left-hand-side (LHS) functions of the constraints **must be linear functions** of the decision variables.

1. The World Health Council is devoted to improving health care in the underdeveloped countries of the world. It now has five medical teams available to allocate among three such countries to improve their medical care, health education, and training programs. Therefore, the council needs to determine how many teams (if any) to allocate to each of these countries to maximize the total effectiveness of the five teams. The teams must be kept intact, so the number allocated to each country must be an integer.

   The measure of performance being used is additional person-years of life. Table 1 gives the estimated additional person-years of life (in multiple of 1000) for each country for each possible allocation of medical teams.

   Table 1 Data for the World Health Council problem

Allocated	Additional Person-Years of Life (in thousands)		
Medical Teams	Country 1	Country 2	Country 3
0	0	0	0
1	45	20	50
2	70	45	70
3	90	75	80
4	105	110	100
5	120	150	130

   a) Formulate the problem as a finite state finite horizon problem and solve it using dynamic programming (DP) algorithm.
   Note: Clearly define and denote the key elements of the deterministic finite state finite horizon problem, that is, planning horizon, decision epochs (or stages), state variables, decision variables (or controls), transition function (or system equation), cost (or contribution for maximization) function, and objective function.

   b) Formulate the problem as an integer program (IP) and solve it using Gurobi.
   Note: Clearly define the notation and explain the IP model.

Submission requirements:

1. For each (sub)problem, name the solution file as "*problem_x.ext*," where "*x*" represents the (sub)problem number (x = 1, 2, 3 or x = 1a, 1b, 1c) and the file extension "*ext*" depends on the file type (Word, Excel, PDF, etc.). If the solution to a (sub)problem contains multiple files (e.g., a Python package), organize the files into a folder named as "*problem_x*."

2. Note that your Python files must be able to be executed directly. So, use *relative paths*

instead of absolute paths. If necessary, you may provide a short "user manual" on how to execute your codes.

Warning: If the TAs have to manipulate your Python package to verify your solutions, you will be deducted points from your grade.

3. Pack all the "(sub)problem" folders in a zip file named "*hw_##_中文姓名.zip*," where "##" (two digits) represents the homework number, for example, "*hw_09_赵磊.zip*."